

Index notation



1 9

2 7

3

a 9^2

b 9^3

c 9^4

d 9^5

4

a 5^3

b 4^5

c $(-9)^4$

d 7^2

e $12^2 \times 9^3$

f $(-13)^4 \times 5^2$

g $11^2 \times 7^3$

h $3^3 \times (-4)^2$

i $(-6)^3 \times (-2)^2$

j $12^3 \times 15^2$

k 2^6

l $24^3 \times 12$

5

a 2^2

b 3^3

6

a 64

b -27

c 243

d 128

e 9

f -36

g 68

h 130

i 140

j 98

k -37

l 27

Expanded notation

7

a 9×9

b $9 \times 9 \times 9$

c $9 \times 9 \times 9 \times 9$

d $9 \times 9 \times 9 \times 9 \times 9$

8

Words	Expanded notation	Index notation
Six to the power of four	$6 \times 6 \times 6 \times 6$	6^4
Eight cubed	$8 \times 8 \times 8$	8^3
Eleven to the power of three	$11 \times 11 \times 11$	11^3
Twenty three to the power of five	$23 \times 23 \times 23 \times 23 \times 23$	23^5
Seven squared	7×7	7^2
Fifteen to the power of four	$15 \times 15 \times 15 \times 15$	15^4

9

a 5×5

c $4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 9 \times 9$



b $7 \times 7 \times 7 \times 7 \times 7 \times 6 \times 6 \times 6 \times 6$

d $77 \times 77 \times 77 \times 77 \times 77 \times 13 \times 13$

10

a $5 \times 5 \times 5$

b 125